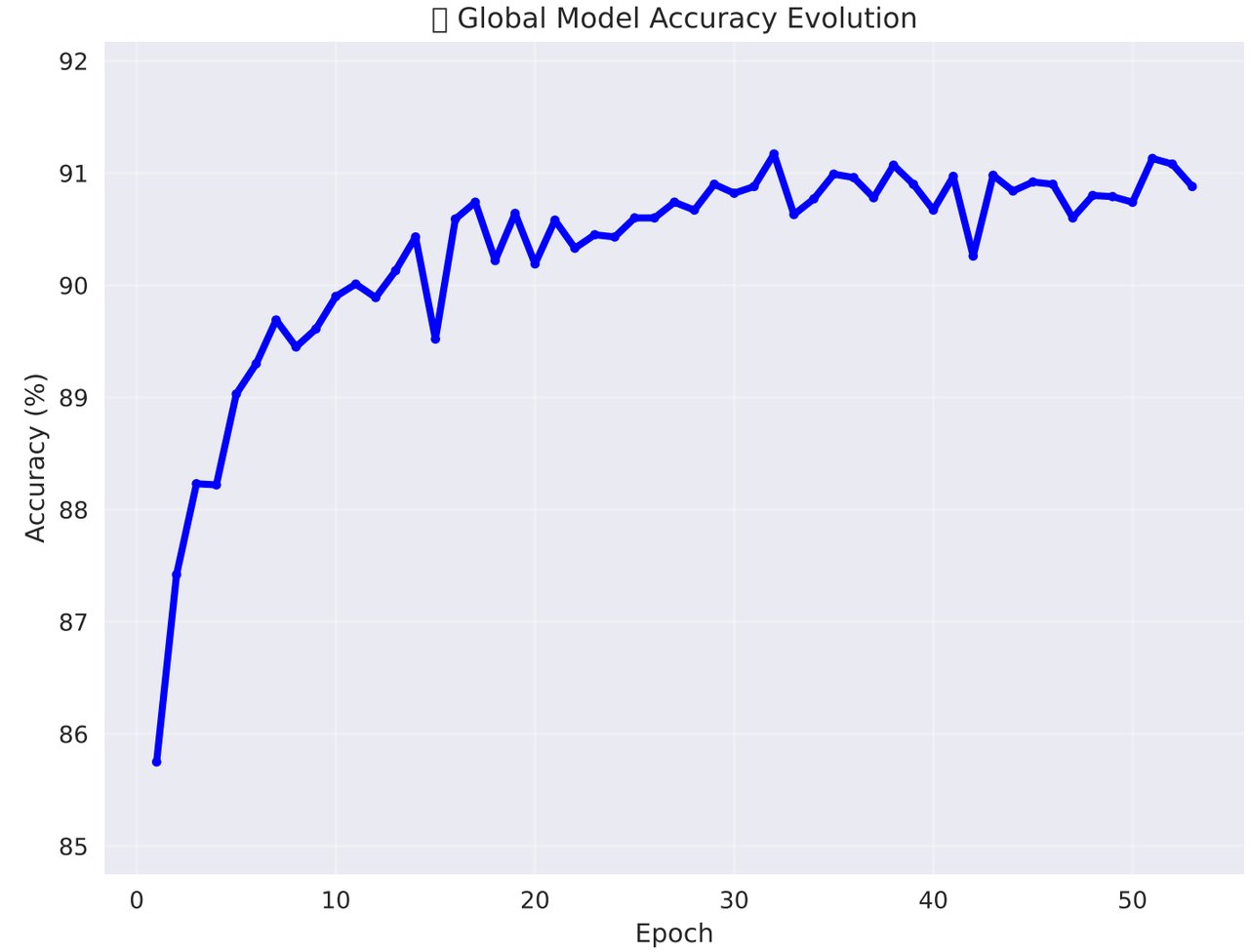
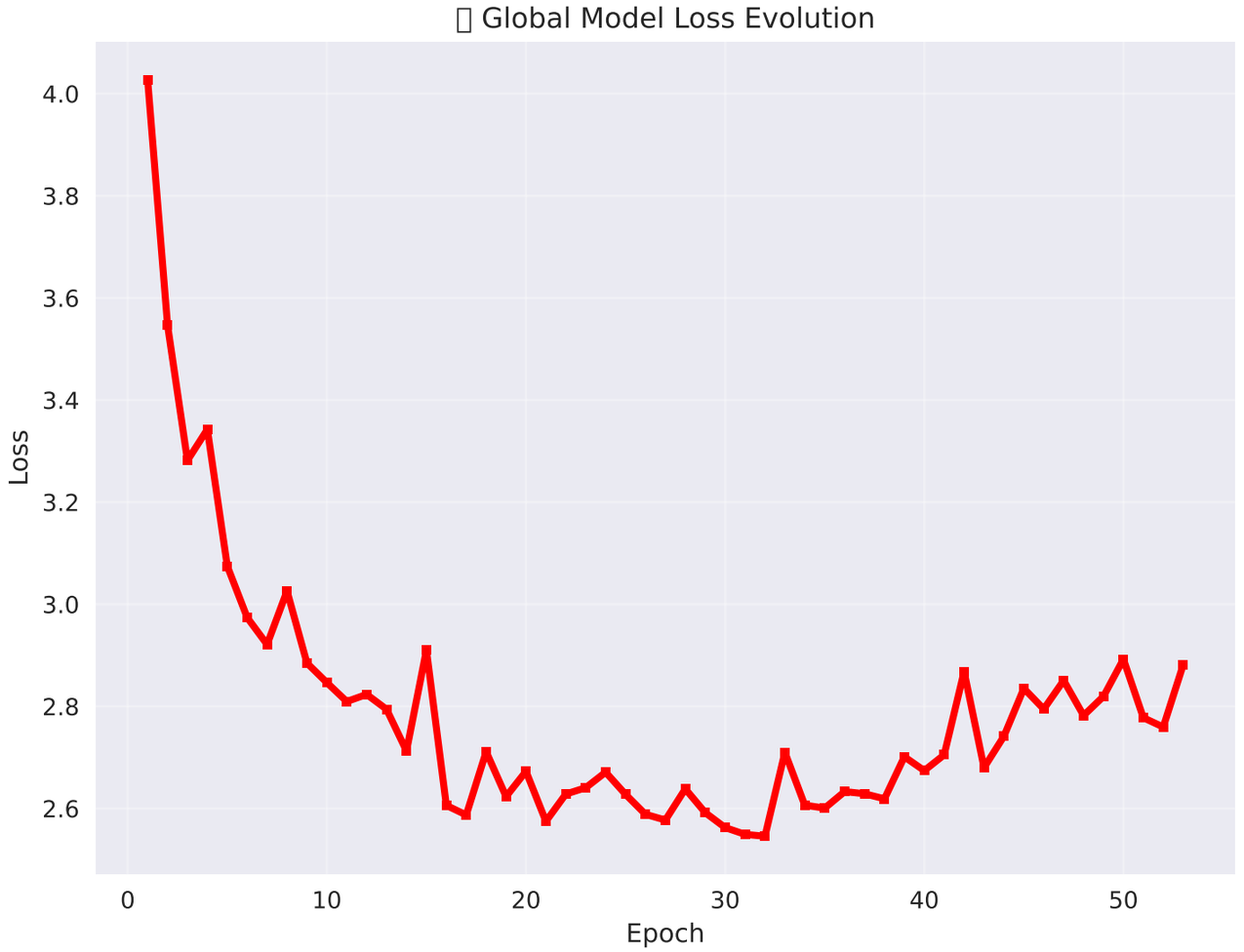
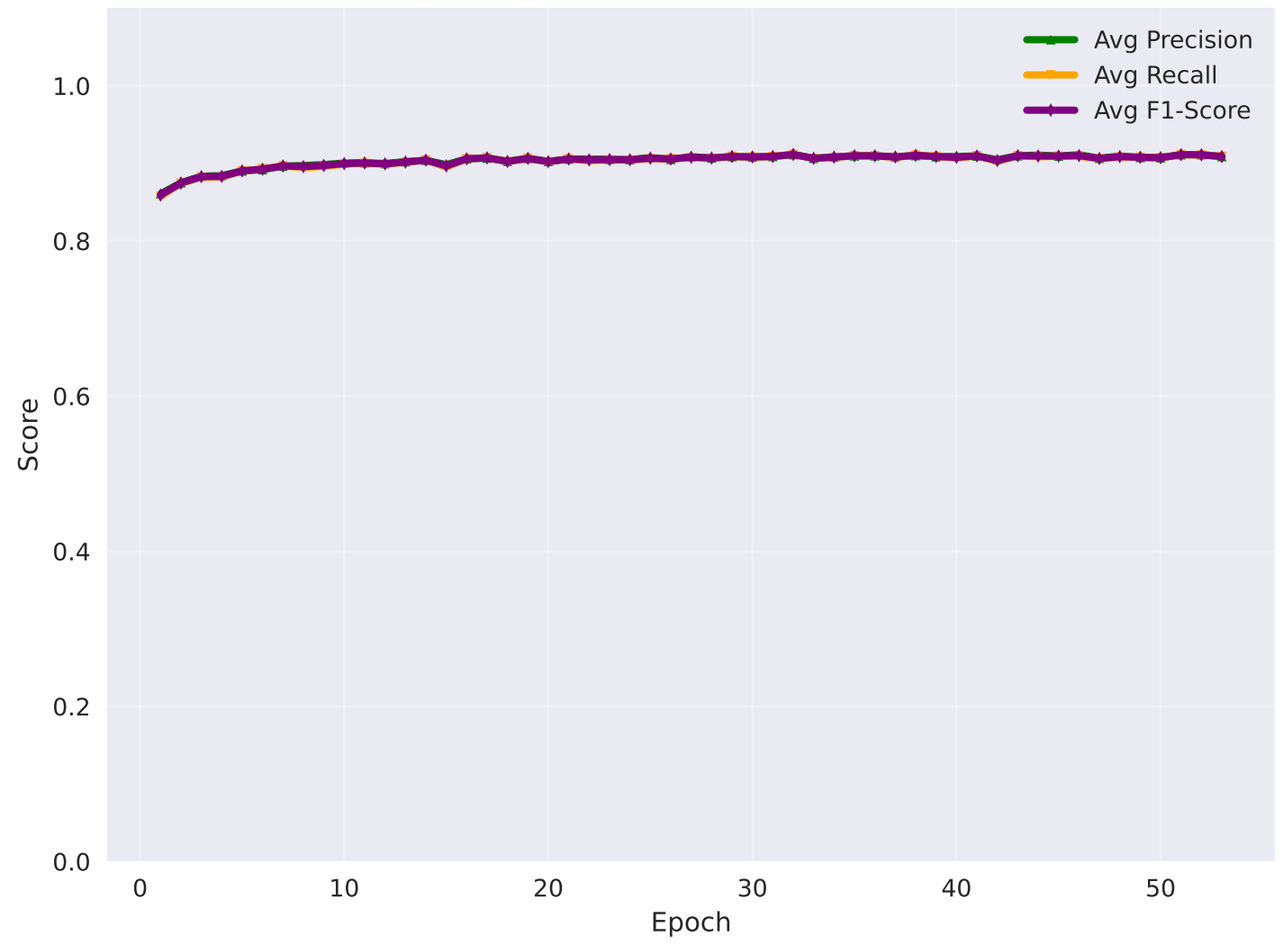
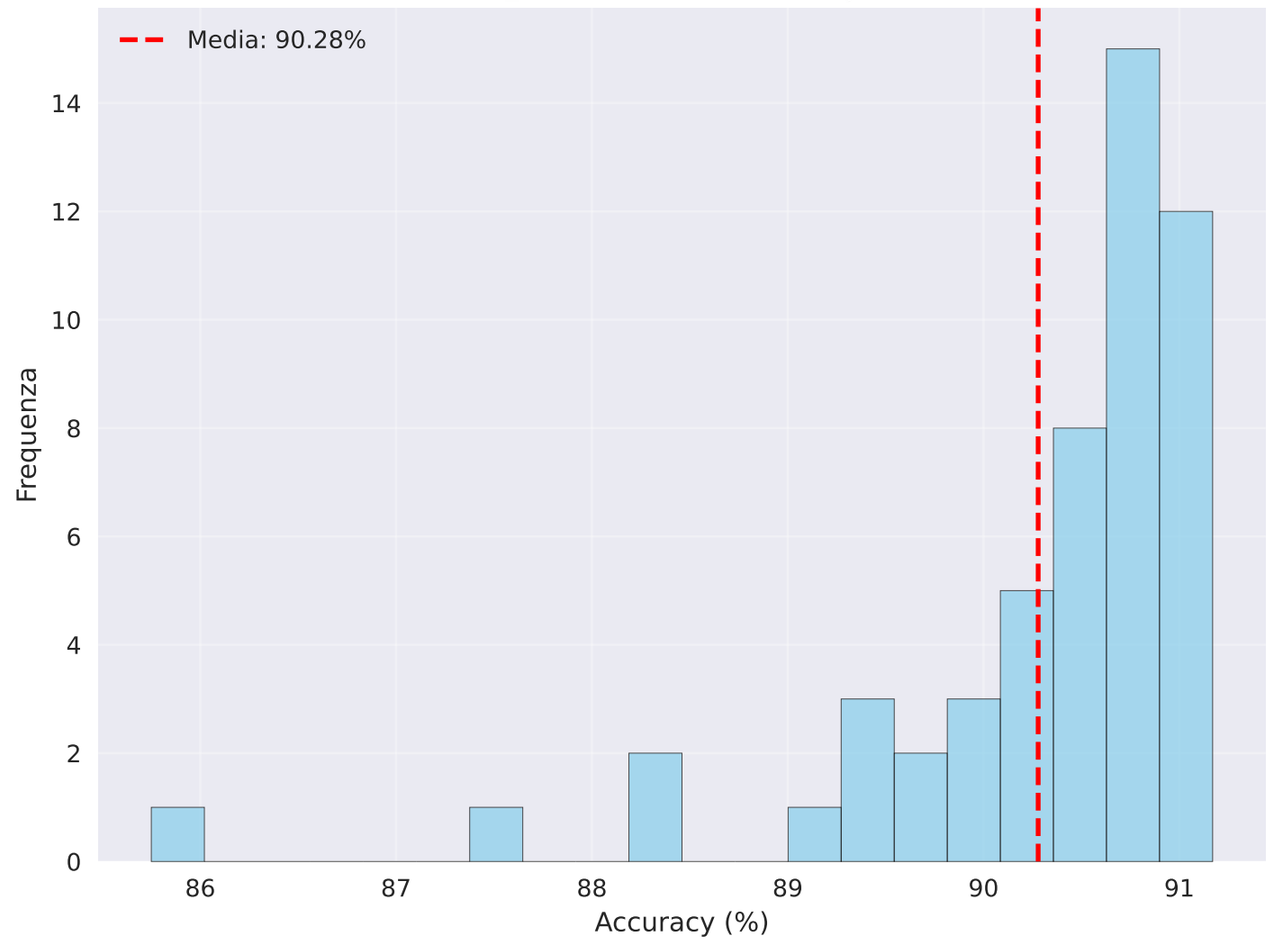
Federated Learning - Main Metrics Overview

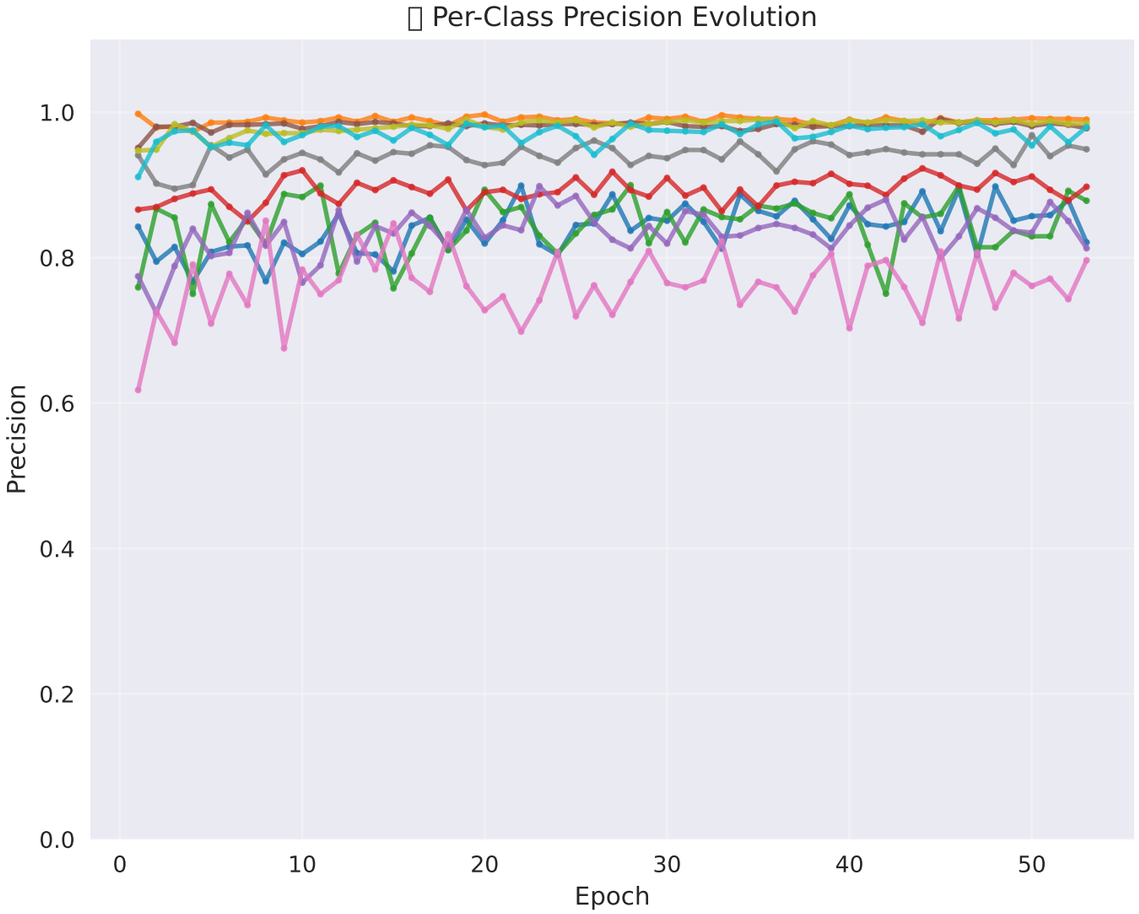




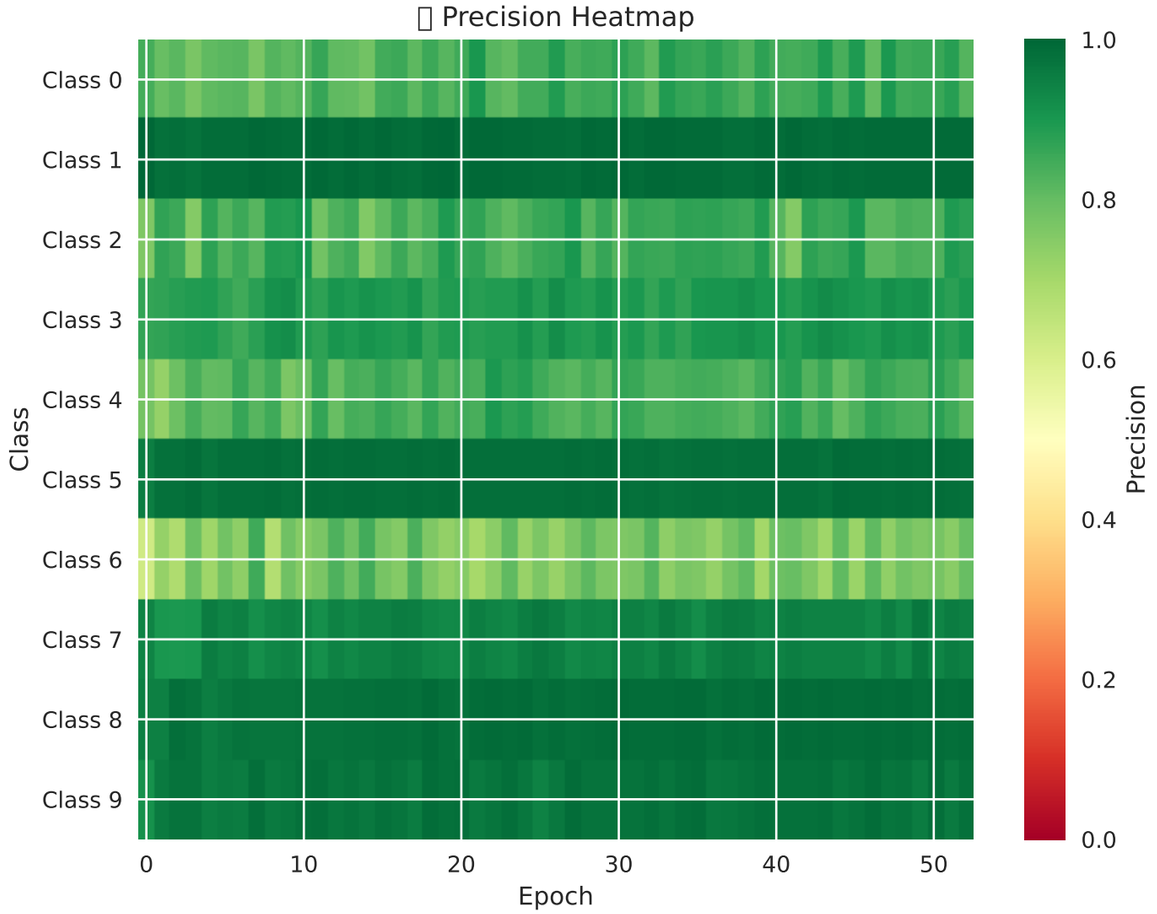


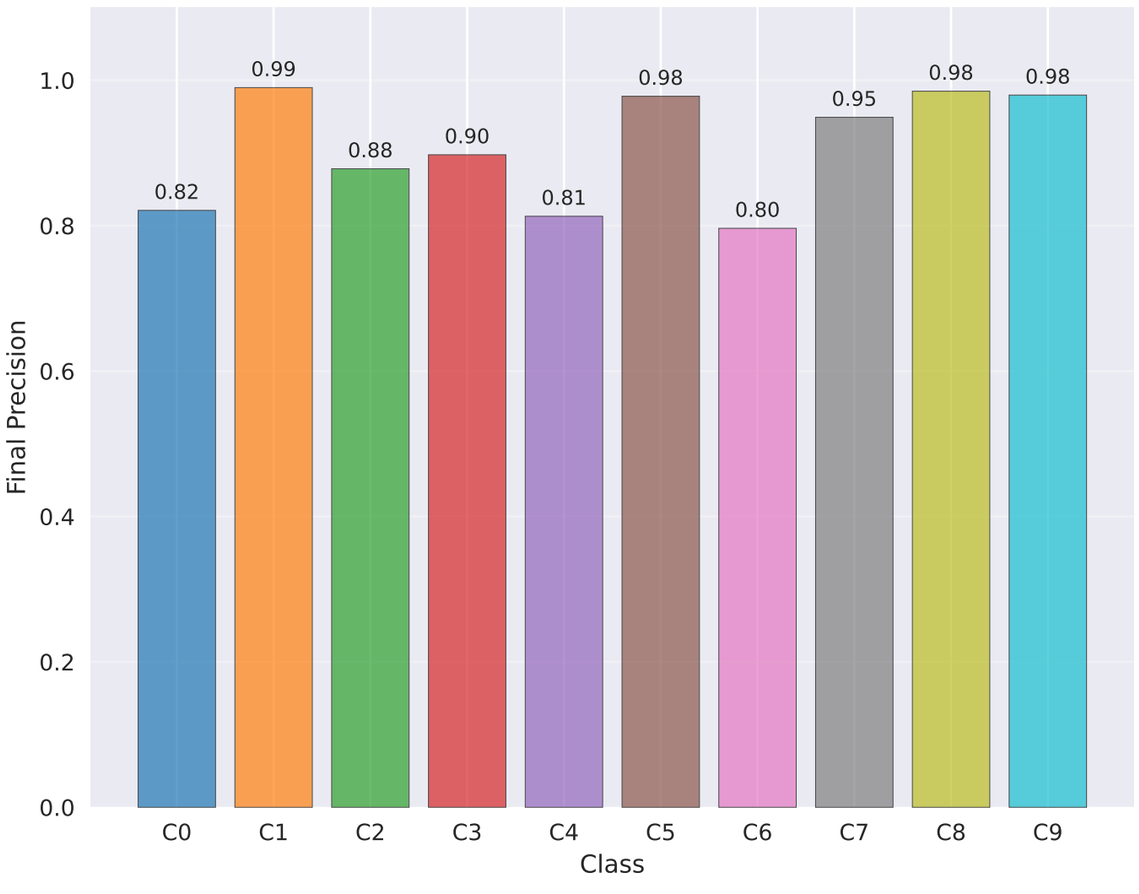


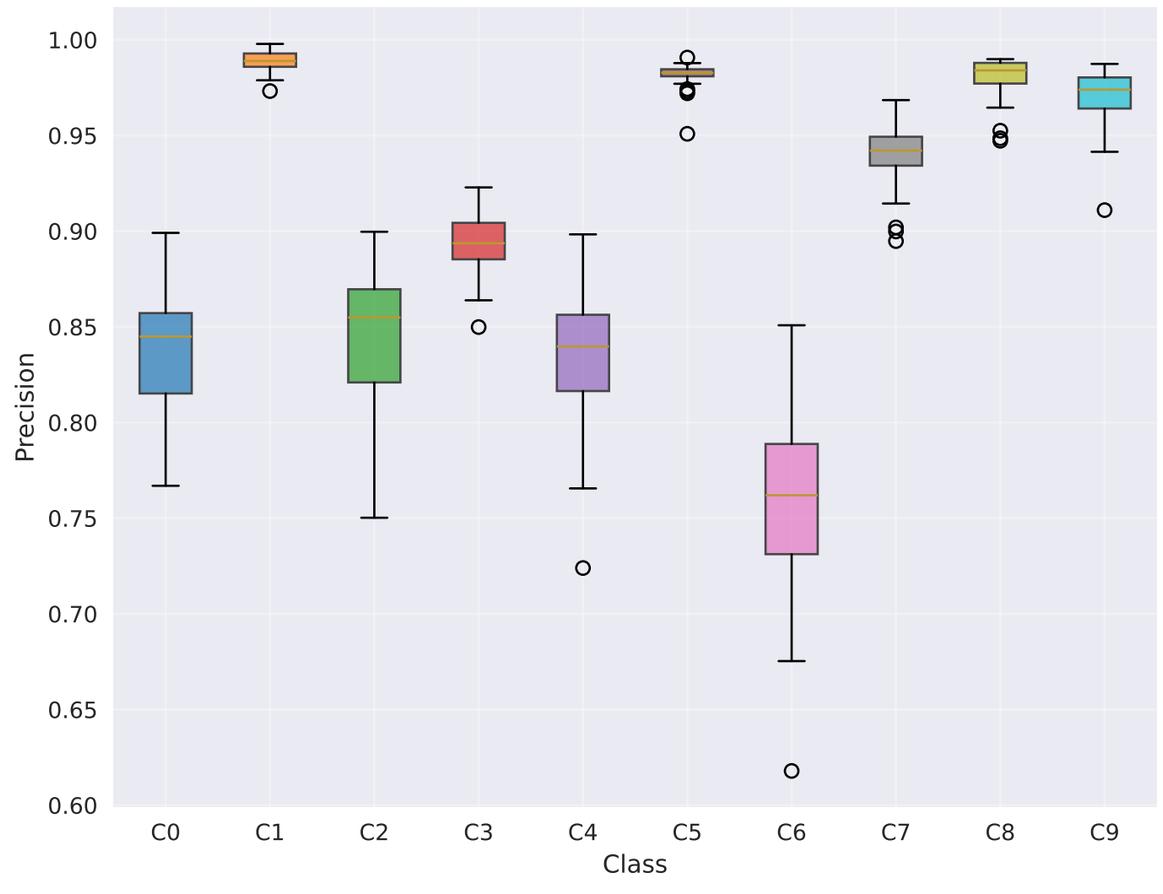
Per-Class Precision Analysis





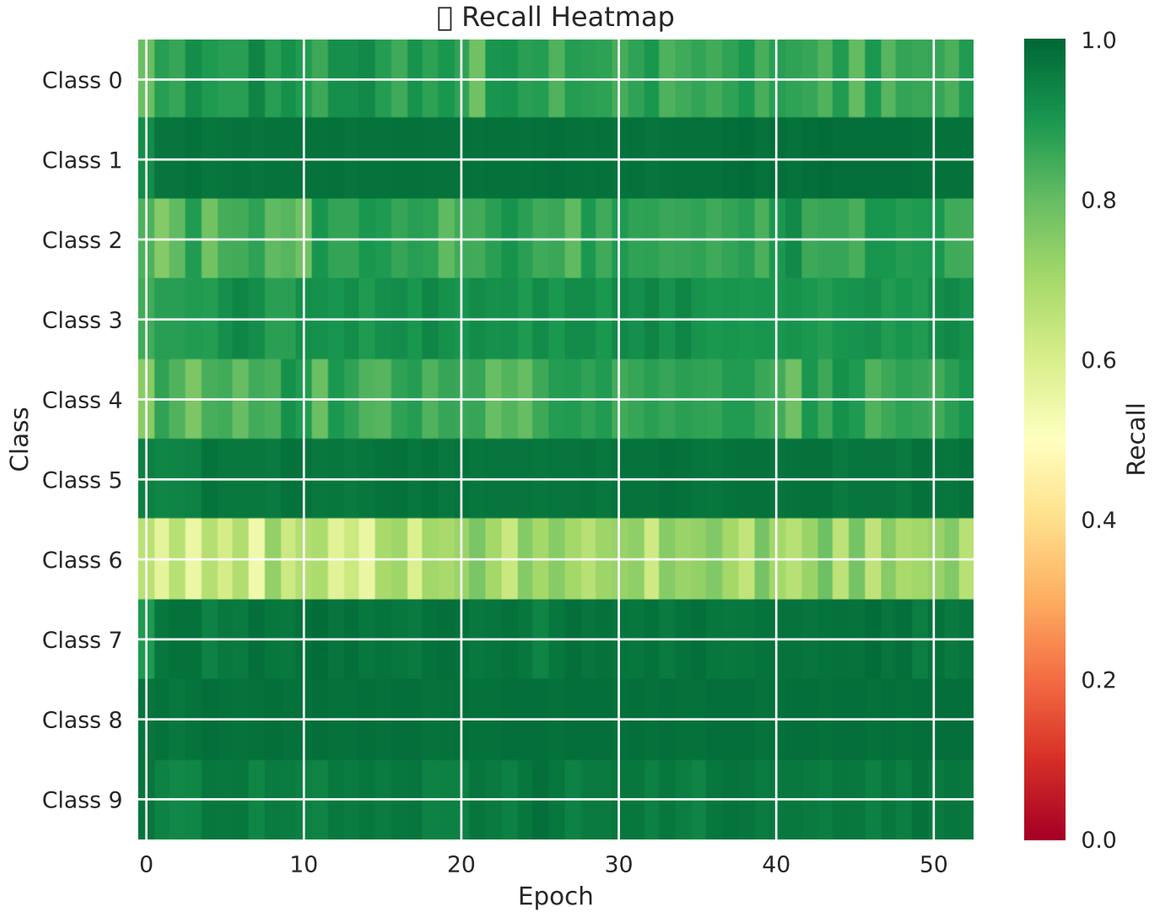


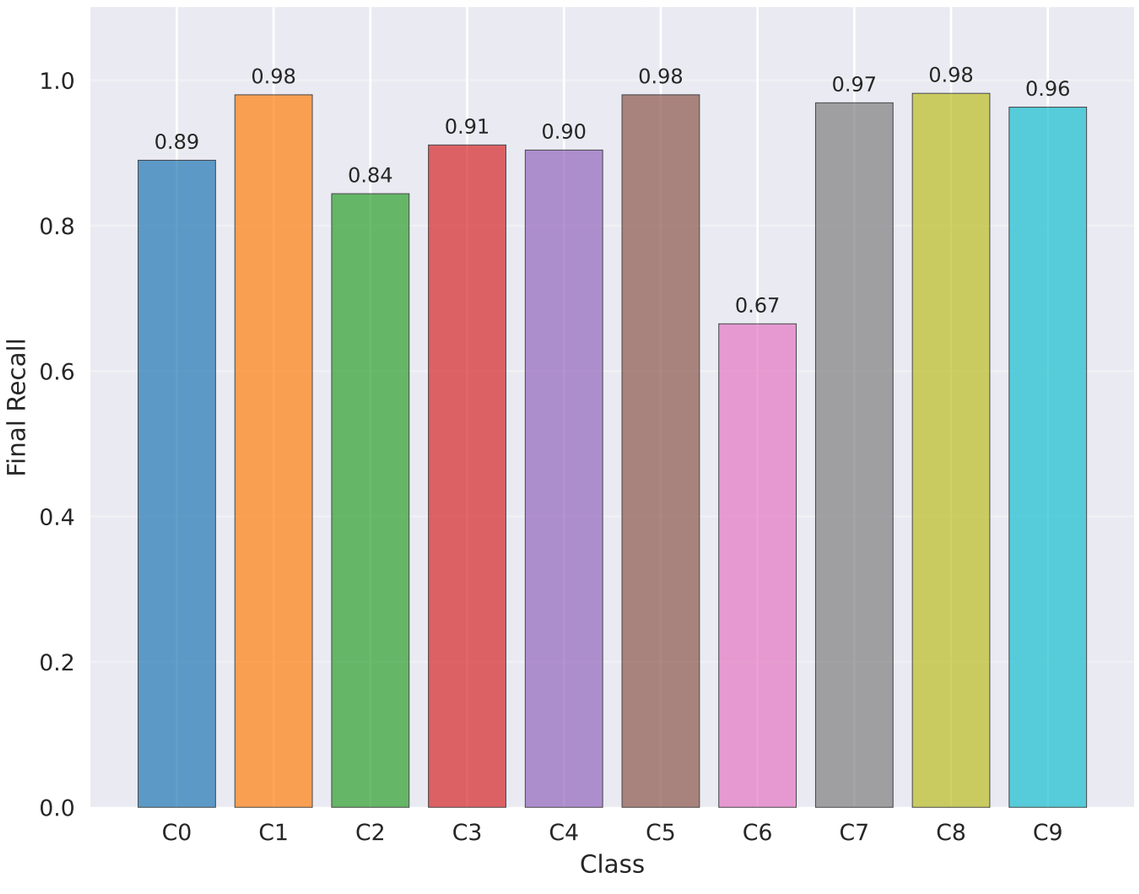


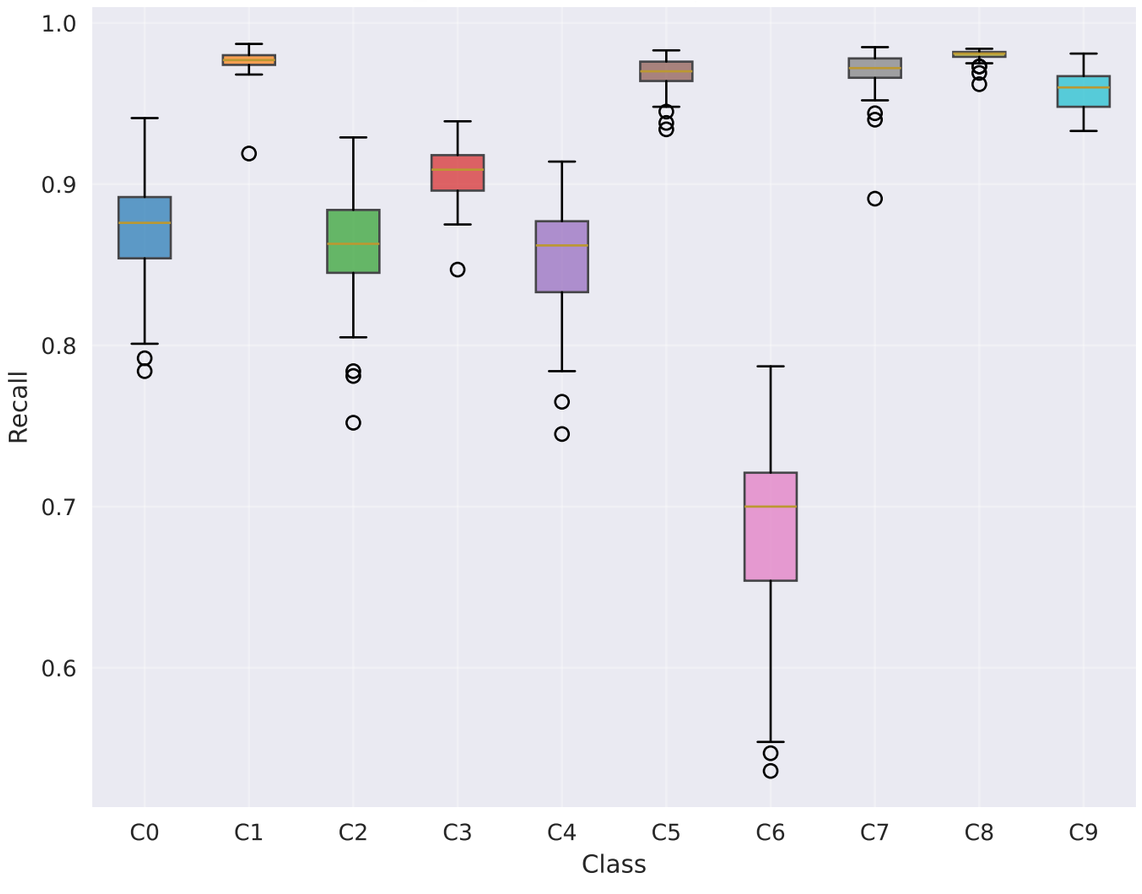


Per-Class Recall Analysis

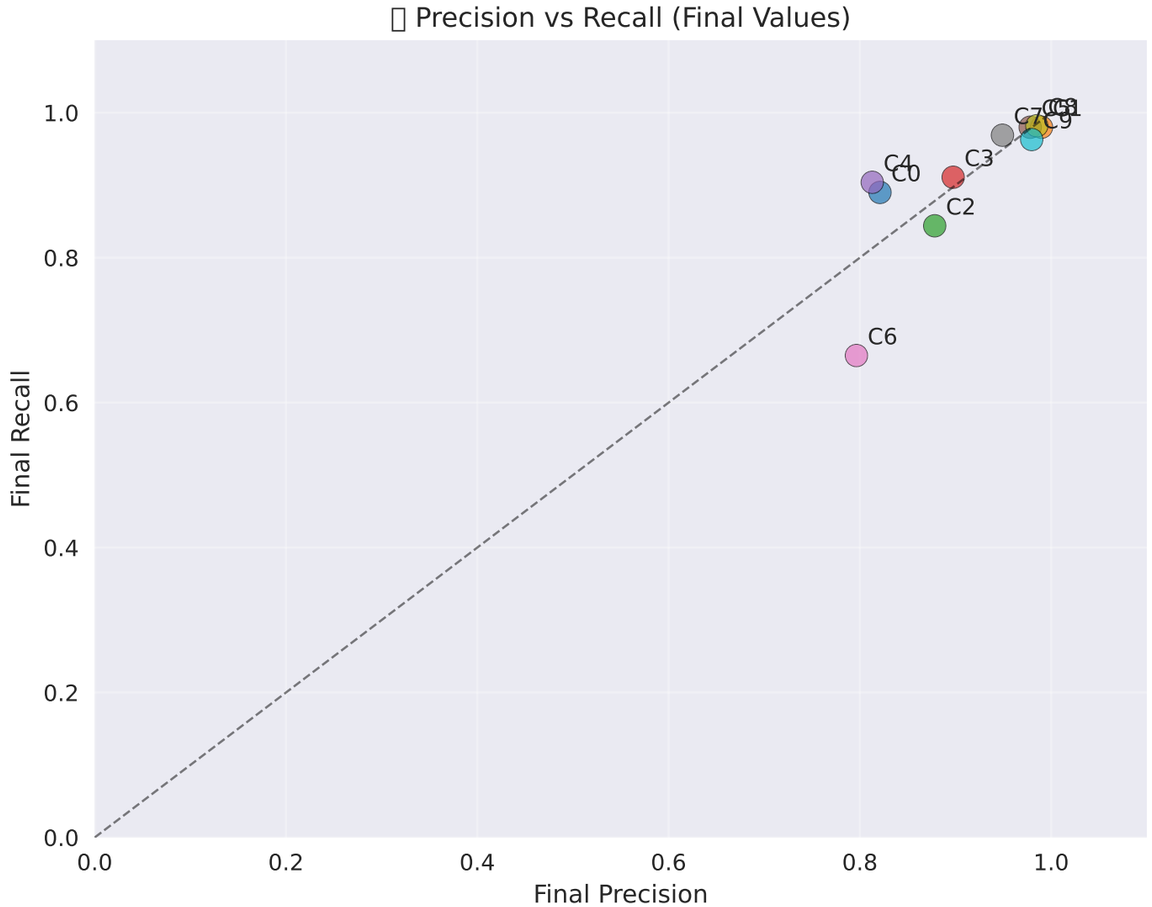


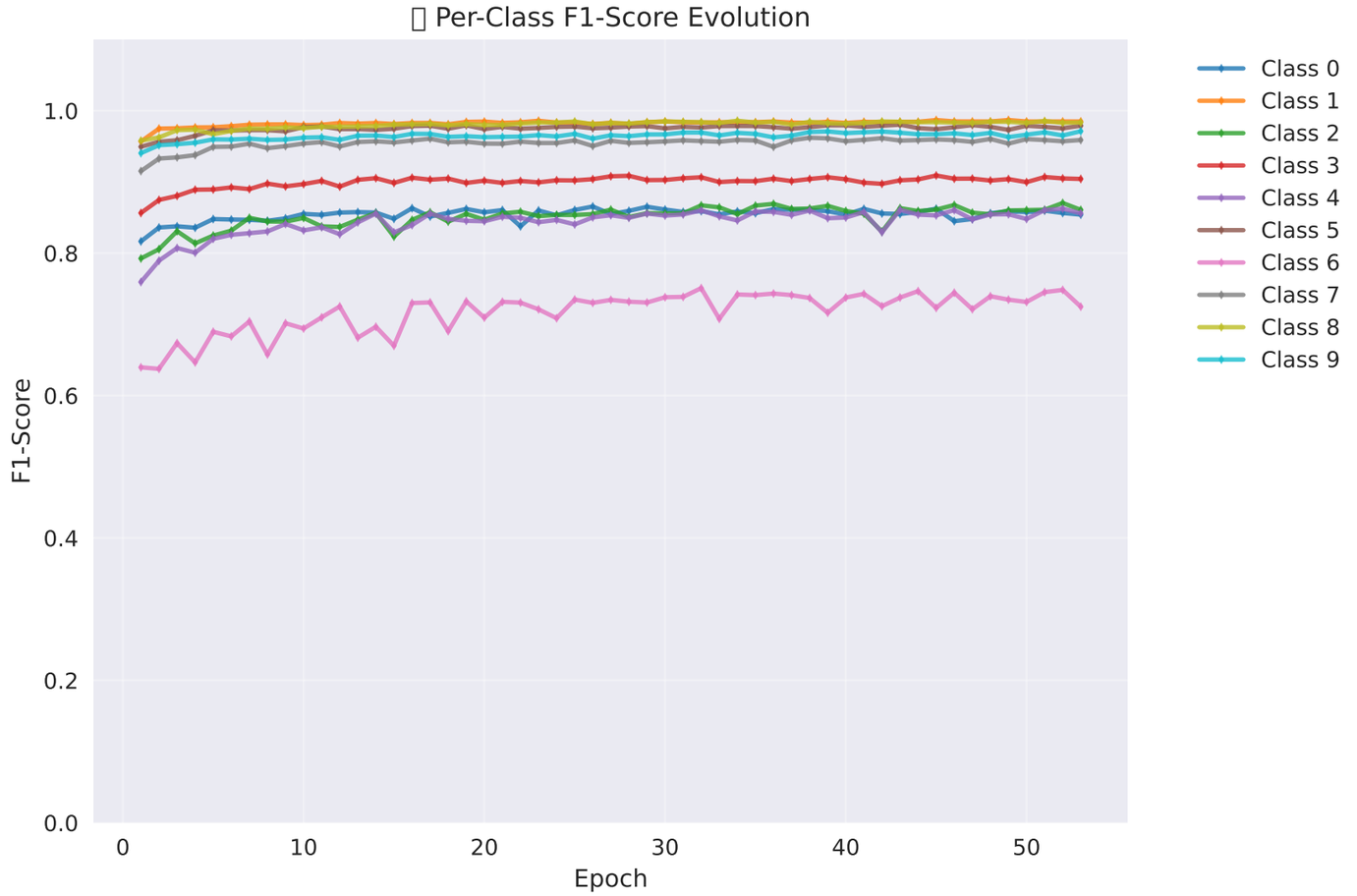


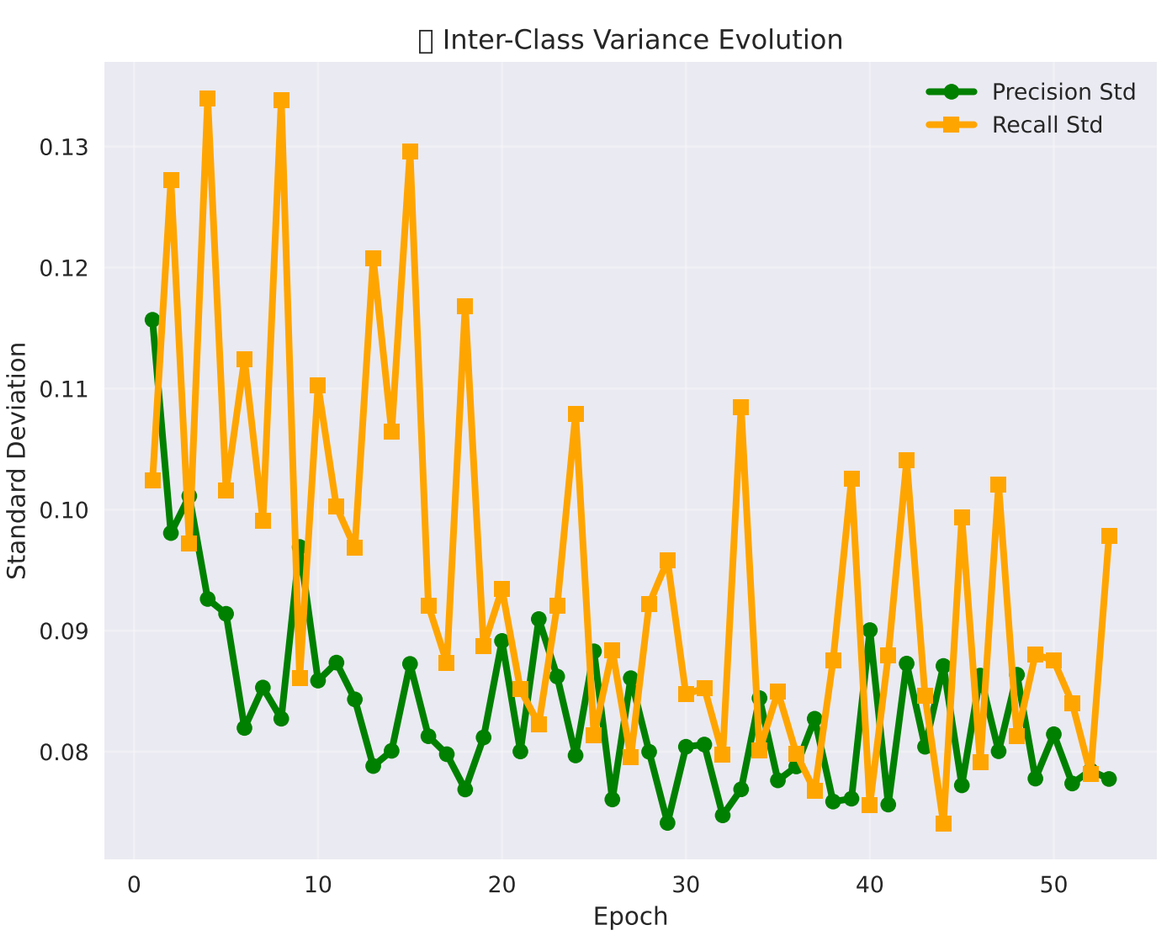


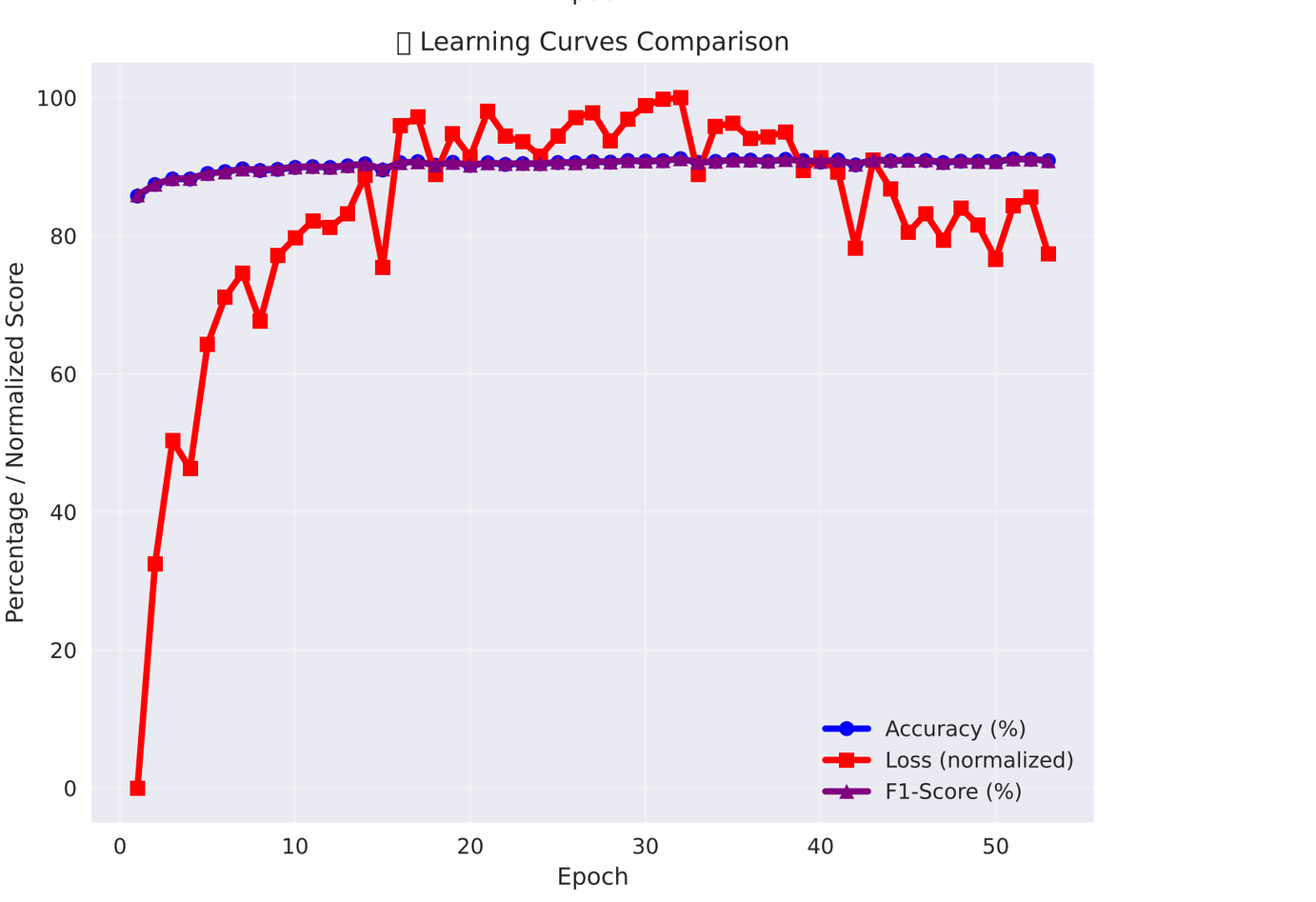


Comparative Analysis









Performance Summary Report

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FEDERATED LEARNING - PERFORMANCE SUMMARY
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FILE ANALIZZATO: 3000_results.csv
DATA ANALISI: 2025-07-28 15:07:22
EPOCHE TOTALI: 53

METRICHE FINALI:
• Accuracy: 90.88%
• Loss: 2.8812
• Average Precision: 0.909
• Average Recall: 0.909
• Average F1-Score: 0.909

EVOLUZIONE TRAINING:
• Miglioramento Accuracy: +5.13%
• Riduzione Loss: +1.1463
• Stabilità Accuracy (CV): 0.0111

PERFORMANCE PER CLASSE:
• Migliore Precision: Classe 1 (0.990)
• Peggiore Precision: Classe 6 (0.796)
• Migliore Recall: Classe 8 (0.982)
• Peggiore Recall: Classe 6 (0.665)

ANALISI DETTAGLIATA PER CLASSE:
• Classe 0: P=0.821, R=0.890, F1=0.854
• Classe 1: P=0.990, R=0.980, F1=0.985
• Classe 2: P=0.878, R=0.844, F1=0.861
• Classe 3: P=0.898, R=0.911, F1=0.904
• Classe 4: P=0.813, R=0.904, F1=0.856
• Classe 5: P=0.978, R=0.980, F1=0.979
• Classe 6: P=0.796, R=0.665, F1=0.725
• Classe 7: P=0.949, R=0.969, F1=0.959
• Classe 8: P=0.985, R=0.982, F1=0.983
• Classe 9: P=0.980, R=0.963, F1=0.971

VALUTAZIONE QUALITATIVA:
• Performance Globale: BUONA
• Stabilità Training: STABILE
• Balance Classes: BILANCIATO

RACCOMANDAZIONI:
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